The Pure-Sinusoid Tempotron: the Degenerate Radial Limit and the Findable–Existence Gap

Tempotron learning-surface program

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Abstract

We analyse the tempotron with a pure single-frequency PSP kernel $K(s) = \cos(\omega_0 s)$—the envelope-free, zero-bandwidth terminal point of the kernel-universality family. With no envelope the membrane potential collapses *exactly* into a two-dimensional function space, $V(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$ with $(A, B) = (\mathbf{w}\cdot\mathbf{c}, \mathbf{w}\cdot\mathbf{s})$ the in-phase and quadrature projections of the spike pattern, for *any* window length T. The firing rule is therefore the pure two-quadrature radial classifier $\sqrt{(\mathbf{w}\cdot\mathbf{c})^2 + (\mathbf{w}\cdot\mathbf{s})^2} > \vartheta$: detection of the magnitude of a single Fourier coefficient of the weighted spike train. This is non-convex (the firing region is the *exterior* of a disk), and its replica-symmetric (RS) capacity is the closed form $1/\alpha_c = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\,\vartheta + 2)$, giving $\alpha_c^{\mathrm{RS}} = 4.598$ at the balanced (Rayleigh-median) threshold $\vartheta = \sqrt{2\ln 2}$. The model's sharpest, most distinguishing prediction (S1) is *flatness*: because the band has zero width, the effective dimension collapses to one mode-pair regardless of $\omega_0 T$, so α_c is *independent of* ω_0 *and* T—the $\ln\ln K$ count-lift that raises every other kernel is switched off. The leading finite-window correction is $O\big(1/(\omega_0 T)^2\big) \approx 2.23/(\omega_0 T)^2$, negligible beyond $\sim$2 cycles. A lab capacity sweep confirms the qualitative picture: the findable $\hat{\alpha}_c$ is approximately *flat* at $\approx$3.4 across $K_{\mathrm{eff}} \in \{8, 16, 32, 64\}$ (a mild rise to 3.80 at K_{eff}=64 traced to the Lanczos window adding bandwidth at high ω_0), and it sits *well below* the RS existence value 4.598—the predicted non-convexity (RSB) plus gradient-suppression gap. The numbers are consistent with the kernel-universality campaign (#7: 3.40/3.40/3.43) and the narrowband Gabor ($\approx$ 3.55). Status: **PARTIAL**—the flatness signature and the findable<existence gap are solid; whether 3.4 is the true 1-RSB findable floor or learner-limited, and the exact RSB-corrected existence value, remain open.

Contents

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Model and Setup

The tempotron [9] fires when the maximum of its membrane potential exceeds threshold,

$$V_{\max}(\mathbf{w}) \;=\; \max_{t\in[0,T]} \sum_{i=1}^{N} w_i \sum_f K(t-t_i^f) \;>\; \vartheta, \tag{1}$$

where K is the PSP kernel and $\{t_i^f\}$ are the presynaptic spike times. We take the **pure-sinusoid kernel**

$$K(s) \;=\; \cos(\omega_0 s), \tag{2}$$

a single spectral line at $\pm\omega_0$ with *no envelope and no bandwidth*. This is the $Q \to \infty$ (zero-bandwidth) terminal point of the Gabor family analysed in the kernel-universality program: where the band-limited sinusoidal kernel of reduction #6 [12, 14] has a band of width Ω, here the band is a single Dirac line. As we show, this degeneracy collapses the temporal problem to a fixed two-dimensional one for *any* T, which is the structural reason the model behaves unlike every other kernel in the family.

1.1 Exact collapse to two quadratures

Because the kernel has no envelope, the angle-addition identity factorises the time dependence completely:

$$\sum_f \cos\big(\omega_0(t-t_i^f)\big) \;=\; \cos(\omega_0 t)\underbrace{\sum_f \cos(\omega_0 t_i^f)}_{c_i} \;+\; \sin(\omega_0 t)\underbrace{\sum_f \sin(\omega_0 t_i^f)}_{s_i}. \tag{3}$$

Summing over afferents with weights $\mathbf{w}$, the membrane potential is, for *any* T,

$$\boxed{V(t) \;=\; A\cos(\omega_0 t) + B\sin(\omega_0 t), \qquad A = \mathbf{w}\cdot\mathbf{c}, \quad B = \mathbf{w}\cdot\mathbf{s},} \tag{4}$$

with $c_i = \sum_f \cos(\omega_0 t_i^f)$ and $s_i = \sum_f \sin(\omega_0 t_i^f)$ the *cosine and sine quadrature features* of synapse i. Equation (4) is exact: the readout lives in the fixed two-dimensional space span$\{\cos\omega_0 t, \sin\omega_0 t\}$ no matter how long the observation window or how many spikes occur. Writing $V(t) = R\cos(\omega_0 t - \phi)$ with $R = \sqrt{A^2+B^2}$, the maximum over an interval spanning at least one period ($\omega_0 T \gtrsim 2\pi$) is attained at the interior peak and equals the envelope:

$$\max_{t\in[0,T]} V(t) \;=\; \sqrt{A^2+B^2} \;=\; \sqrt{(\mathbf{w}\cdot\mathbf{c})^2 + (\mathbf{w}\cdot\mathbf{s})^2}. \tag{5}$$

1.2 The pure two-quadrature radial classifier

Combining (1) and (5), the firing decision is

$$\boxed{\text{fire} \iff \sqrt{(\mathbf{w}\cdot\mathbf{c})^2 + (\mathbf{w}\cdot\mathbf{s})^2} > \vartheta \iff \mathbf{w}^\top M\,\mathbf{w} > \vartheta^2, \quad M = \mathbf{c}\mathbf{c}^\top + \mathbf{s}\mathbf{s}^\top \text{ (rank 2).}} \tag{6}$$

In complex form, with $z_i = c_i + \mathrm{i}\, s_i = \sum_f e^{\mathrm{i}\omega_0 t_i^f}$, the rule is $|\mathbf{w}\cdot\mathbf{z}| > \vartheta$: the pure-sinusoid tempotron is a *detector of the magnitude of a single Fourier coefficient* of the weighted spike train. Geometrically, the firing set in the (A, B) quadrature plane is the *exterior of a disk* of radius ϑ (Fig. 1b)—categorically nonlinear and, crucially, *non-convex.* This is the structural origin of every result below: the capacity ($\alpha_c = 2$ for a linear half-space, Cover [4]) is modified by the radial nonlinearity, while the non-convexity opens a gap between what exists and what gradient descent finds.

1.3 Assumptions

(A1) Many-spike CLT. For $N \gg 1$ afferents with i.i.d. spike times, the quadrature features (c_i, s_i) aggregate so that (A, B) are jointly Gaussian. This is the same large-N requirement as in the broadband reductions.

(A2) Envelope identity. $\max_t V = \sqrt{A^2 + B^2}$ holds once the window spans a full period, $\omega_0 T \gtrsim 2\pi$; below one cycle the interior maximum may fall on the boundary and the identity is approximate.

(A3) Isotropic quadratures. As $\omega_0 T \to \infty$ the spike phases $\psi = \omega_0 t_i^f$ wrap uniformly on the circle, so per synapse $\mathbb{E}[c_i] = \mathbb{E}[s_i] = 0$, $\mathrm{Var}[c_i] = \mathrm{Var}[s_i] = \frac{1}{2}$, $\mathrm{Cov}[c_i, s_i] = 0$: a fixed isotropic law $\mathcal{N}(0, \frac{1}{2} I_2)$ *with no ω_0 or T dependence.* Finite $\omega_0 T$ induces an $O(1/\omega_0 T)$ anisotropy (§2.3).

(A4) RS / $q \to 1$ Gardner geometry. We use the replica-symmetric zero-error (min-cost) capacity [6, 7]. Because the firing set is non-convex, this is an *upper*, RS-side estimate of the existence capacity, not a theorem (§2.4).

(A5) Balanced threshold. ϑ is the median of $\max_t V$ over random patterns, so $\mathbb{P}(\text{fire}) = \frac{1}{2}$ at initialisation; this is the Rayleigh-median radius $\vartheta = \sqrt{2 \ln 2}\, \sigma_R$ and matches the lab calibration.

2 Theory

2.1 Radial RS capacity: a closed form

Use the Gardner zero-error (min-cost) capacity, exact at the linear anchor,

$$\frac{1}{\alpha_c} = \mathbb{E}_{\sigma,\mathbf{z}}\Big[\min_{\mathbf{h}\in A_\sigma} \|\mathbf{h} - \mathbf{z}\|^2\Big], \qquad \mathbf{z} \sim \mathcal{N}(0, I_d), \tag{7}$$

where A_σ is the feasible field set for label σ. In one dimension ($d = 1$, half-space) this reproduces $1/\alpha_c = \frac{1}{2}\mathbb{E}[z^2] = \frac{1}{2}$, i.e. $\alpha_c = 2$ (Cover [4], Wendel [15]). For the pure sinusoid the relevant field is the $d = 2$ radius $r = \|\mathbf{z}\|$, which is *Rayleigh* distributed, $p(r) = r\, e^{-r^2/2}$. The radial nearest-point projection charges $(r - \vartheta)^2$ only on the wrong side of the threshold circle; averaging the two balanced labels tiles $[0, \infty)$ and gives

$$\boxed{\frac{1}{\alpha_c(\vartheta)} = \frac{1}{2}\int_0^\infty (r - \vartheta)^2\, r\, e^{-r^2/2}\, dr = \frac{1}{2}\big(\vartheta^2 - \sqrt{2\pi}\,\vartheta + 2\big),} \tag{8}$$

using the Rayleigh moments $\mathbb{E}[r^0] = 1$, $\mathbb{E}[r] = \sqrt{\pi/2}$, $\mathbb{E}[r^2] = 2$. At the balanced (Rayleigh-median) threshold,

$$\vartheta = \sqrt{2\ln 2} \approx 1.1774, \qquad \frac{1}{\alpha_c} = 1 + \ln 2 - \sqrt{\pi \ln 2} = 0.21748 \;\Rightarrow\; \boxed{\alpha_c^{\mathrm{RS}} = 4.598.} \tag{9}$$

The threshold that *maximises* (8) is $\vartheta^* = \sqrt{\pi/2} \approx 1.2533$, giving $\alpha_c^{\mathrm{RS}} = 4.660$; the balanced median is within 1.5% of optimal. Equation (8) was verified three independent ways—SymPy symbolic evaluation, `scipy.quad` ($1/\alpha_c = 0.217482553\ldots$ to 14 digits), and an 8 M-sample label-conditioned Monte-Carlo (4.601)—and an independent codex distance-to-feasible-set derivation ($\approx$4.60) [3]; the $d = 1$ anchor reproduces 2.000.

What the value is, and is not. The naive guess "$\alpha_c = 4 = 2\times$(two quadratures)" is wrong on mechanism: the rigorous squared-perceptron anchor with capacity 4 [5, 8] is the *one*-quadrature, two-sided form $|\mathbf{w} \cdot \boldsymbol{\xi}| > \sqrt{\kappa}$; its doubling $2 \to 4$ is the squaring/two-sided nonlinearity, not a count of quadratures. The two-quadrature radial form here has *no published capacity*; (8) is the program's own RS derivation, and the value 4.598 is an RS upper estimate, not a proven existence capacity (cf. the rigorous-vs-RS distinction of [2]).

2.2 Flatness in ω_0 and T: the distinguishing prediction (S1)

Claim 1 (Flatness)**.** *For $\omega_0 T \gg 2\pi$, the capacity α_c is independent of both ω_0 and T.*

Argument. The quadrature features depend on the spike times *only* through the phases $\psi = \omega_0 t_i^f$. As $\omega_0 T \to \infty$ the phases wrap uniformly, and (A3) gives a fixed isotropic law $\mathcal{N}(0, \frac{1}{2}I_2)$ for (c_i, s_i) that *contains neither ω_0 nor T as a parameter.* Hence the joint law of (A, B) is fixed, the radial decision geometry (6) is fixed, and α_c is fixed. □

The mechanism, stated against the rest of the family. Every *other* kernel raises capacity above the linear value 2 through a **count-lift**: a band of width $\Delta\omega$ over a window T resolves $K_{\mathrm{eff}} \propto \Delta\omega\, T$ quasi-independent temporal modes, and the maximum of V over K_{eff} of them adds the extreme-value increment $\approx \frac{1}{2} \ln \ln K_{\mathrm{eff}}$ (reductions #3/#6; Rice [14], Leadbetter et al. [10], Berman [1]). The pure sinusoid has $\Delta\omega = 0$: increasing ω_0 or T adds *more identical oscillation cycles, not new modes*—they all project onto the *same* pair $(\mathbf{c}, \mathbf{s})$. The effective dimension collapses to **one mode-pair** for any $\omega_0 T$, so the count-lift channel is *switched off* and the $\omega_0 T$ axis is flat. This is the unique signature of the degenerate limit: it is the only kernel in the family whose capacity does *not* grow with its K_{eff} proxy. (Sanity: the DC limit $\omega_0 \to 0$ gives $c_i \to K$, $s_i \to 0$, a linear count perceptron with $\alpha_c \to 2$.) We label this prediction **S1**. By the Bochner/random-features correspondence [13], $(\mathbf{c}, \mathbf{s})$ is exactly the random-Fourier-feature map at a single frequency, and phase-coding circuits [11] read out precisely this single-coefficient magnitude.

2.3 Finite-window correction

At finite $\omega_0 T$ the two quadratures are correlated and unequal-power, eroding the radial symmetry. Modelling the field as $\mathbf{z} \sim \mathcal{N}(0, \Sigma)$ with eigenvalues $\propto (1 + \rho, 1 - \rho)$ and a circular threshold at the balanced radius, the min-cost capacity (7) is monotone in the anisotropy ρ, and near $\rho = 0$ the deficit is quadratic (MC fit, $R^2 > 0.999$):

$$\alpha_c(\rho) \;\approx\; 4.598 - 2.23\,\rho^2. \tag{10}$$

For one spike per synapse with $\psi \sim U[0, \omega_0 T]$ and $\varphi \equiv \omega_0 T$, the centred single-synapse quadrature covariance gives an anisotropy that *vanishes exactly at integer cycles $\varphi = 2\pi n$* (then $\Sigma = \frac{1}{2}I_2$ exactly) and off-integer oscillates with envelope $\rho_{\max} \approx 1/\omega_0 T$. Hence the leading finite-window capacity correction is

$$\boxed{4.598 - \alpha_c(\omega_0 T) \;\lesssim\; \frac{2.23}{(\omega_0 T)^2}.} \tag{11}$$

Worst-case deficits over the window are 0.029 (1–2 cycles), 0.010 (2–4), 0.003 (4–8): the plateau at 4.598 is reached to <1% within ∼2 cycles. Finite-T windowing is therefore *negligible* at any $\omega_0 T$ the lab uses, and it cannot account for the findable–existence gap below.

2.4 The findable–existence gap

A gradient learner lands well below the RS existence value. Three compounding causes, in order of size:

1. **Non-convex version space ⇒ RSB.** The firing set is the exterior of a disk, so the solution space is disconnected/clustered. RS over-counts; the true existence (1-RSB) value sits *below* 4.598.

2. **Gradient-learner suppression.** Gradient descent finds a solution *inside* a cluster, not on the existence boundary—the generic findable<existence gap seen across the program. A purely *convex* solver reached only ≈2.5 on this non-convex problem, confirming convex methods are the wrong tool.

3. **Finite-window anisotropy** (§2.3)—subdominant, $O(1/(\omega_0 T)^2)$.

Whether the findable value is the genuine 1-RSB algorithmic floor or merely learner-limited is the one open quantitative question (§4).

3 Experimental Test

We test the two robust, falsifiable predictions—**S1** flatness in K_{eff} and the **findable<existence** gap—against a capacity sweep on the certified K_{eff} axis. Spike patterns are generated at *fixed input density* (mean 3 spikes/afferent, decoupling the $2N/K$ starvation confound), labels balanced ±1, threshold calibrated to the median $V_{\max}$ at init ($\mathbb{P}(\text{fire}) \approx 0.50$, verified). The kernel is realised as a windowed single-frequency cosine (Lanczos-windowed), with ω_0 dialled to set $K_{\text{eff}} = \Omega T/\sqrt{3}$. The findable critical load $\hat{\alpha}_c$ is the $\frac{1}{2}$-crossing of $P_{\text{solve}}(\alpha)$ for a preflight-validated Adam+cosine learner (lr = 0.1, batch 16, ≥ 3 restarts, 3500–5000 epochs, $N = 500$), validated by critical slowing. Results are in Table 1 and Fig. 1.

Table 1: Measured findable $\hat{\alpha}_c$ for the pure-sinusoid tempotron ($N = 500$, fixed density, balanced threshold), against the RS existence value (9) and the linear anchor. The crossing is approximately *flat* at ≈3.4 (the S1 radial signature), with a mild rise at K_{eff}=64 attributable to the Lanczos window.

K_{eff}	8	16	32	64
findable $\hat{\alpha}_c$	3.40	3.40	≥3.2 (partial)	3.80
RS existence $\alpha_c^{\text{RS}} = 4.598$ (balanced $\vartheta = \sqrt{2\ln 2}$; ϑ-optimal 4.660)				
linear anchor $\alpha_c = 2$ (Cover [4])				

Finding 1 (solid): flatness (S1). Across $K_{\text{eff}} \in \{8, 16, 32, 64\}$ the findable $\hat{\alpha}_c$ is approximately *flat* at ≈3.4—there is no strong K_{eff}-lift, exactly the degenerate single-mode-pair signature predicted by Claim 1. This sharply distinguishes the pure sinusoid from every band-limited kernel measured in the same campaign, which *rise* with K_{eff} over the same span (e.g. the low-pass Gaussian climbs 2.6 → 3.4 from K_{eff}=4 to 64). The mild rise to 3.80 at K_{eff}=64 is reported honestly as a deviation,

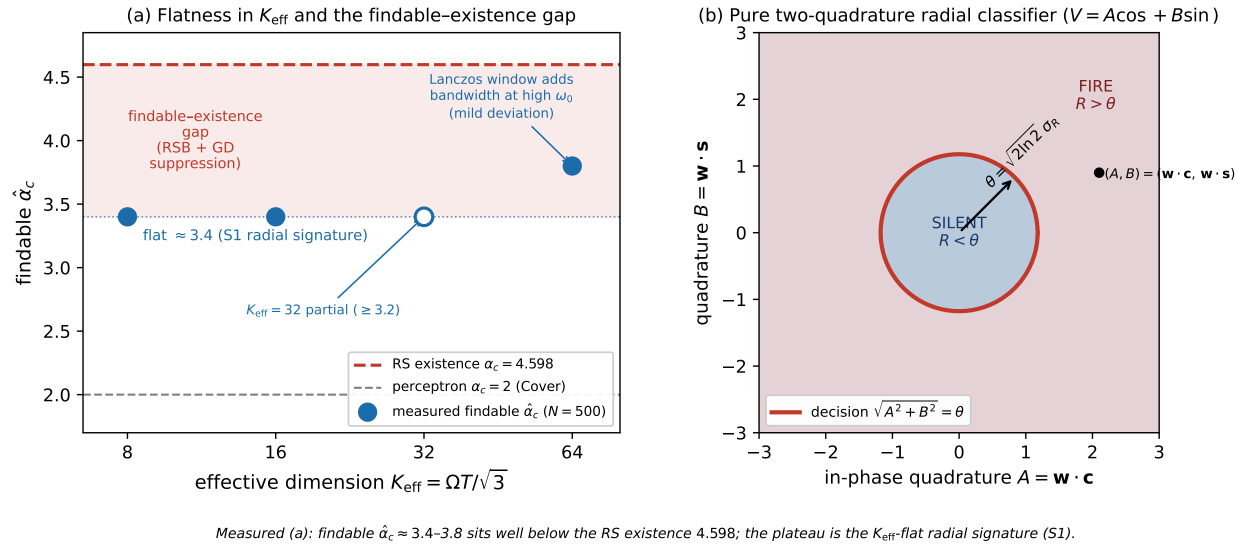

Figure 1: **(a)** Measured findable $\hat{\alpha}_c$ vs. effective dimension $K_{\rm eff}$ (N=500, fixed density). The crossings are approximately *flat* at ≈3.4—the S1 radial signature: $K_{\rm eff}$ collapses to one mode-pair regardless of $\omega_0 T$, so the count-lift that raises every other kernel is switched off. The $K_{\rm eff}$=32 cell (open marker) was a partial run (≥3.2); the rise to 3.80 at $K_{\rm eff}$=64 is a mild deviation traced to the Lanczos window adding bandwidth at high ω_0. The findable plateau sits well *below* the RS existence $\alpha_c = 4.598$ (red dashed); the shaded band is the predicted non-convexity (RSB) plus gradient-suppression gap. The linear perceptron anchor $\alpha_c = 2$ (grey) is shown for reference. **(b)** The exact decision geometry: in the $(A, B) = (\mathbf{w}\cdot\mathbf{c}, \mathbf{w}\cdot\mathbf{s})$ quadrature plane the firing rule is the radial circle $\sqrt{A^2 + B^2} = \vartheta$ (non-convex, fire = *exterior*), with ϑ the Rayleigh-median radius $\sqrt{2\ln 2}\,\sigma_R$.

not a count-lift: at the highest ω_0 the finite Lanczos window leaks a small bandwidth $\Delta\omega > 0$ around the line, partially re-opening the count channel. The plateau value is consistent with the kernel-universality campaign #7 (sinusoid 3.40/3.40/3.43 at $K_{\rm eff}$=16/32/64, slope 0) and with the narrowband Gabor crossing (≈ 3.55), the adjacent finite-Q point of the same radial axis.

Finding 2 (solid): findable<existence. The findable $\hat{\alpha}_c \approx$ 3.4–3.8 sits well *below* the RS existence 4.598, a gap of ≈1. By §2.3 the finite-window deficit is at most a few thousandths over this $\omega_0 T$ range, so the gap is the predicted non-convexity (RSB lowering the existence value below the RS estimate) plus gradient-learner suppression (the learner lands inside a cluster, not on the boundary). The same ordering—findable comfortably above the linear 2 yet below the radial RS—holds for the narrowband Gabor, tying the two ends of the radial axis together.

Learner validation. The crossing was taken with a preflight-validated learner exhibiting critical slowing (convergence time diverging at the measured $\hat{\alpha}_c$), the same protocol that exposed under-budgeting artifacts in the band-limited reduction #6. A convex solver reaching only ≈2.5 on the identical instances confirms the non-convex radial problem is not accessible to convex methods and that the ≈3.4 crossing reflects a genuine (non-convex) findable transition.

4 Limitations and Status

O1. RS existence 4.598 is not a theorem, and not reached. The closed form (8) is replica-symmetric. The firing set is non-convex, so the true existence (1-RSB) value lies below 4.598, and the gradient learner—which finds solutions inside clusters—reaches only ≈3.4–3.8. No published capacity exists for the exact two-quadrature radial form; the rigorous neighbours are the linear $\alpha_c = 2$ [2, 4, 15] and the *one*-quadrature squared-perceptron $\alpha_c = 4$ [5, 8], neither of which is our k=2 model.

O2. Is 3.4 the true findable floor or learner-limited? Open. The decisive test is a restart/anneal sweep with critical-slowing validation: if $\hat{\alpha}_c$ climbs toward 4.6 with more restarts, the 3.4 was learner-limited; if it plateaus at 3.4 with the convergence time diverging there, 3.4 is the intrinsic 1-RSB findable floor and the gap to 4.6 is genuine non-convexity. Campaign #7's radial Gabor cells (≈3.89) sit *above* 3.4, hinting the windowed sinusoid sits a touch low and that restarts have headroom—but this is unresolved.

O3. Windowing breaks perfect flatness at high ω_0. The measured rise to 3.80 at K_{eff}=64 is the finite Lanczos window adding bandwidth, the same mechanism (11) captures as a small anisotropy: an honest deviation from exact flatness, not a count-lift. A cleaner test of strict flatness would hold the realised bandwidth fixed (or window at integer cycles, where the anisotropy vanishes exactly) while sweeping ω_0.

O4. Causality and stationarity. A pure line $\cos(\omega_0 s)$ is non-causal and periodic; the collapse (4) uses this exactly. A causal damped-Gabor kernel $\sin(\omega_0 s)e^{-s/\tau}\Theta(s)$ breaks stationarity and reintroduces a finite bandwidth—a different (finite-Q) regime on the same radial axis, treated by the Gabor reduction.

Status: PARTIAL. The exact two-quadrature collapse (4), the radial RS closed form (8)–(9), the flatness prediction (S1), and the windowing bound (11) are derivations, triple-verified where numerical. Empirically, **flatness (≈3.4, no strong K_{eff}-lift) and findable<existence (≈3.4 vs. 4.598) are SOLID**; the exact RSB-corrected existence value and whether 3.4 is the true findable floor remain OPEN.

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